

Project, Program, Portfolio and system

ICT Project Management

Project

A Project is temp. Group activity to design or produce unique product, service or goal.

A project is defined as a specific, finite activity that produces an observable and measurable result under certain preset requirements.

Program

A program is a specific set of ordered operations for a computer to perform.

portfolio

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A portfolio is a compilation of materials that exemplifies your beliefs, skills, qualifications, education, training and experiences. It provides insight into your personality and work ethic.

System

Any specific system aims to support operations, management and decision-making. An information system is the information and communication technology (ICT) that an organization uses, and also the way in which people interact with this technology in support of business processes.

Project objective and Goals

Goals & Objective

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Goals are the agreed outcomes of a project

objectives are the set of operations that need to take place for achieving the project goal.

■ Vision Statement

- Highest Level Statement.
- Shows Direction & Aspiration.
(Which may never be achieved)

■ Goals (Vague)

- High Level Statement.
- Provide overall context for what the project is to achieve.
- Align with biz goals.

■ Objectives (Specific)

- Lower Level Statement.
- Describe desired results
 - Specific, tangible products & deliverables the project will deliver.

Why do we care?

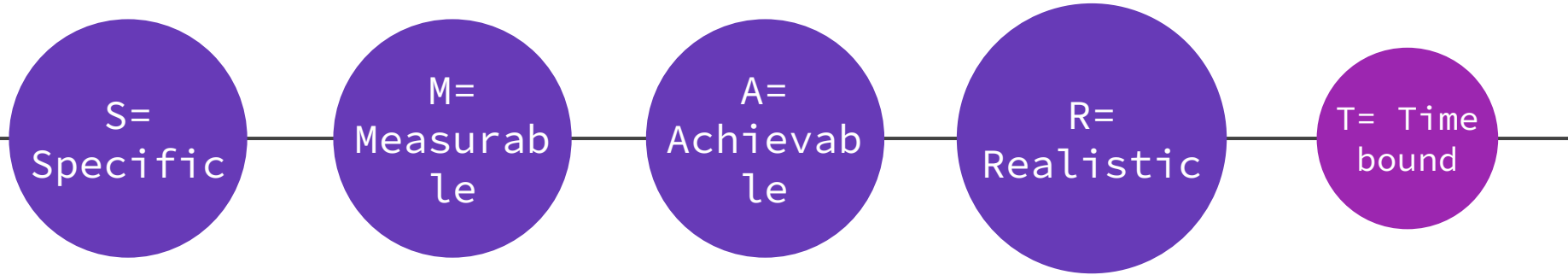
Why do we care?

- Managing a project
Calls for clear Objectives.

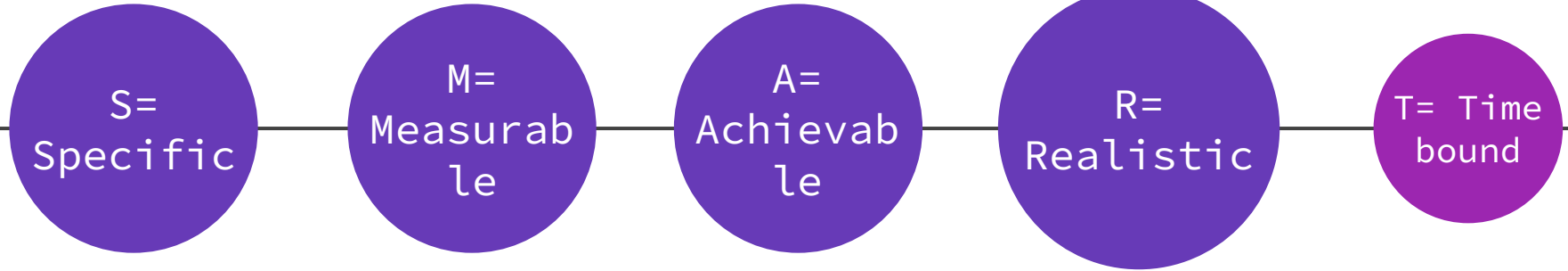
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The more clear,
the more likely to Achieve.

- Used to Structure the Project
and Validate Success.

SMART Goals and Why ?



How to write effective objectives?



A horizontal line with five purple circles of varying sizes. The circles contain the letters S, M, A, R, and T respectively, each followed by an equals sign and a description of the criterion. The circles are connected by a thin horizontal line.

S=
Specific

M=
Measurab
le

A=
Achievab
le

R=
Realistic

T= Time
bound

Example of Goals and Objective:

Goal: Increase overall satisfaction level for customers submitting support tickets through the website.

Objective: Implementing a new ticketing system by Feb 1 to achieve ticket response time no more than 1 hour.

Classification of projects

Based on type of work

Based on completion time

Based on value

On the basis of ownership

On the basis of pace of work

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Classification based on type of work.

Industrial project: Established for manufacturing or trading goods or services.

Non industrial project: These projects are generally initiated by Government. Healthcare, pollution control, Education, Smart city, Traffic control system.

Classification based on completion time.

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Special short duration project: Less than a year

Short duration project: 1-3 years

Medium duration project: 3-5 years

Long duration project: More than 5 years

Classification based on value .

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Small value project: Less than 1 crores

Medium value project: 1-100 crores

Large value project: 100-1000 crores

Mega value project: More than 1000 crores

Classification on the basis of ownership.

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Private sector projects:

Public sector projects:

Government sector projects:

Joint sector projects (ppp):

Classification on the basis of pace of work.

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Normal Track projects:

Fast track projects (crash projects):

Project Constraints

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- **Quality**
- **Time**
- **Cost**
- **Scope**
- **Benefits**
- **Risks**

Quality



Quality

Quality is one of six major constraints of every project, as depicted in the classic **triple constraint triangle**, which also includes scope, time, and cost:

- If you are unable to meet a sudden rise in cost, the project scope may shrink and the quality may decline
- If the project scope extends due to scope creep, you may not have the time or resources to deliver the promised quality
- If delivery time is cut or rushed, project costs may rise and quality will very likely decline

Time

One of the most important stakeholder considerations, project time (how long it will take to deliver), is a vital measure of project success. Your task is to **estimate project time as accurately** as possible, which requires a blend of research and experience.

Cost

Equally important to stakeholders is how much a project will cost. As with time constraints, your budget estimates need to be presented in a range.

- Estimate costs with thoroughly researched market rates for goods and services you need
- Estimate costs with vendor bids and ranges
- hourly cost estimates, be sure to estimate your time accurately.
- Estimate your budget by considering all costs: labor, material, factory, equipment, administrative, software, contractors, etc.
- Look at costs and budgets for similar past projects inside and outside your organization

Scope

Since a project scope is not an estimate but a guaranteed set of deliverables, it's difficult to imagine creating a range for this project constraint. However, you can consider that stakeholders may be invested in scope risk and scope tolerance ranges.

For example, you may list a set of deliverables that could be created if budget and schedule allow, a wish list that your stakeholders can choose from if there's money and time left over after mandatory deliverables are completed.

Benefits

your business case should articulate the project's justification and what set of measures will be used to assess its benefits to the organization.

Risks

We usually think of threats—what might go wrong when we plan for risks. A project manager must be able to reasonably foresee failures at every step of a project, and prepare for them accordingly. This can involve playing out what-if scenarios and formulating contingency plans:

- What if a supplier fails to deliver?
- What if we lose any number of resources due to illness or transfer?
- What if the market takes a huge swing?
- What if our competitor launches a similar product at the same time?

Project management and its advantages

Project management is a powerful business tool that can deliver many advantages to businesses of all sizes.

It gives you repeatable processes, guidelines and techniques to help you manage the people and the work involved in your projects.

It can increase your chances of success and help you deliver projects consistently, efficiently, on time and on budget.

advantages

- improve your chances of achieving the desired result
- gain a fresh perspective on your project, and how it fits with your business strategy
- prioritise your business' resources and ensure their efficient use
- set the scope, schedule and budget accurately from the start
- stay on schedule and keep costs and resources to budget
- improve productivity and quality of work
- encourage consistent communications amongst staff, suppliers and clients
- satisfy the various needs of the project's stakeholders
- mitigate risks of a project failing
- increase customer satisfaction

Project Environment

- Internal Env
- External Env

Internal Environment

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The internal environment of an organization refers to events, factors, people, systems, structures and conditions inside an organization that are generally under the direct control of the company.

- People
- Budget
- Corporate culture

External Environment

The external environment is composed by factors that occur outside the organization but which can cause internal changes and are, for the most part, beyond the company's control.

- Customers,
- competition,
- economy,
- technology,
- political and social conditions,
- resources

Skill Requirement of Project Manager

- Project management is not an easy job.
- In fact, it's several not-easy jobs, including the initiation, planning, executing, controlling, and closing of a project.
- Even more difficult, that project is delegated to a team of your choosing, given specific goals to achieve over a defined timeline for a determined budget.

Skill Requirement of Project Manager

- Leadership
- Communication
- Scheduling
- Risk Management
- Cost Management
- Negotiating
- Critical Thinking
- Task Management
- Quality Management
- A Sense of Humor
- Have Patience
- Tech Gig

Project Management Institute

- The Project Management Institute (PMI) is a global nonprofit professional organization for project management.
- The PMI serves more than 2.9 million professionals, including over 500,000 members in 208 countries and territories around the world.
- PMI offers eight certifications that "recognize knowledge and competency," including the Project Management Professional (PMP)[®] certification, which serves as the standard of professionalism in the field. To earn a PMI credential, candidates must document that they have met required education and experience requirements, then they must pass an exam.

PMBOK

- PMBOK® is the standard by which PMP (Project Management Professional) certification through PMI is obtained.
- PMP is an important industry-recognized certification for project managers.
- The PMBOK® goal is to enable organizations to standardize project management practices across departments.
- It also helps project managers work with a standardized system across organizations.
- This means that a project manager who moves from one company to another can use the same practices.
- The methods are well documented and focus on practices that have been proven to work.
- The guide also discusses what doesn't work, so managers can avoid common mistakes.

Thank you

Computer Science Department, NAST